

COVER PAGE

IGRVT-50 Payload EE

Schematic

Version: 1.0.0



Project: DAL_PDU

Sub-system: IGRVT-50

Design by: KIEN HOANG

Sheet title: CoverPage.SchDoc

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Checked by: BAO BUI

Document No: 1

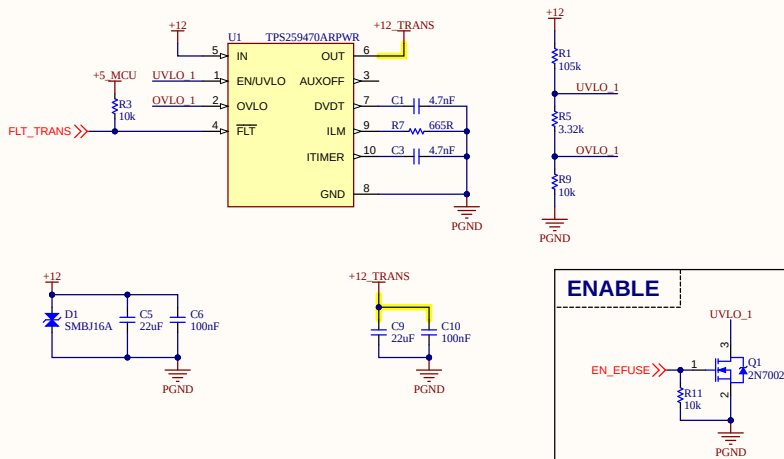
Size: A3

Approved by: VU PHAM

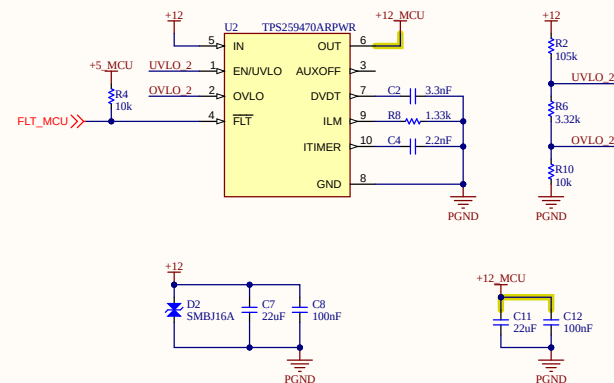
Revision: 1.0.0

Date: 7/10/2026

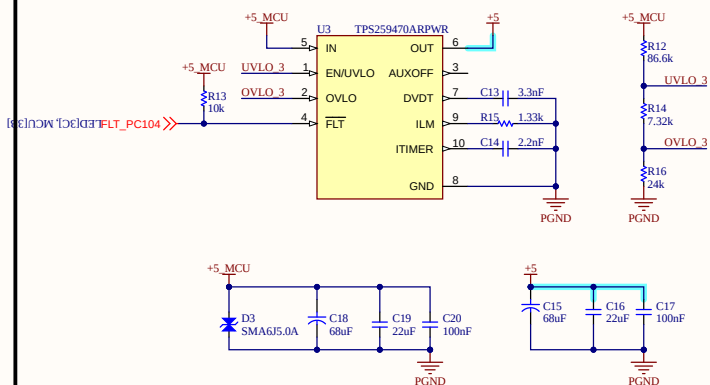
eFuse_Transducer



eFuse_MCU



eFuse_PC104



Project: DAL_PDU

Sub-system: IGRVT-50

Design by: KIEN HOANG

Sheet title: eFuse.SchDoc

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Document No: 2

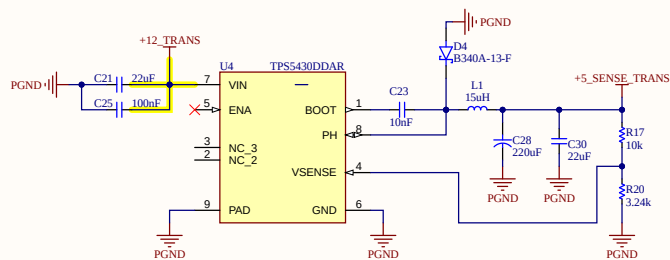
Size: A3

Approved by: VU PHAM

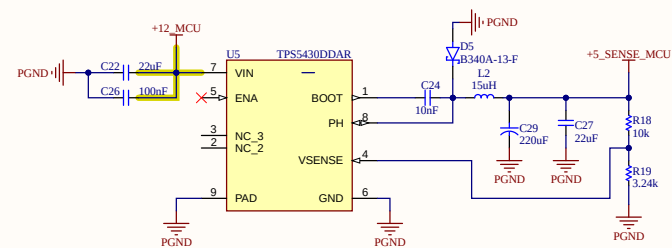
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5V TRANSDUCER



5V MCU



Project: DAL_PDU

Sub-system: IGRVT-50

Design by: KIEN HOANG

Sheet title: Buck 5V.SchDoc

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Document No: 3

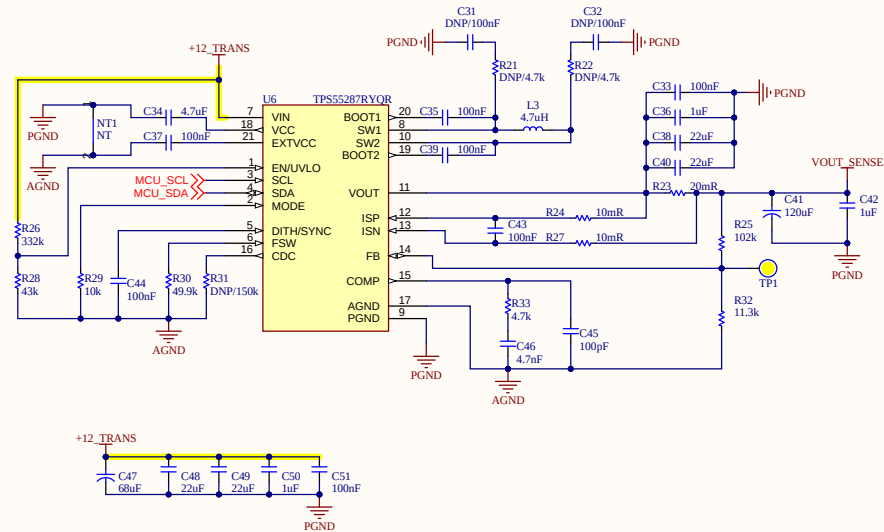
Size: A3

Approved by: VU PHAM

Revision: 1.0.0

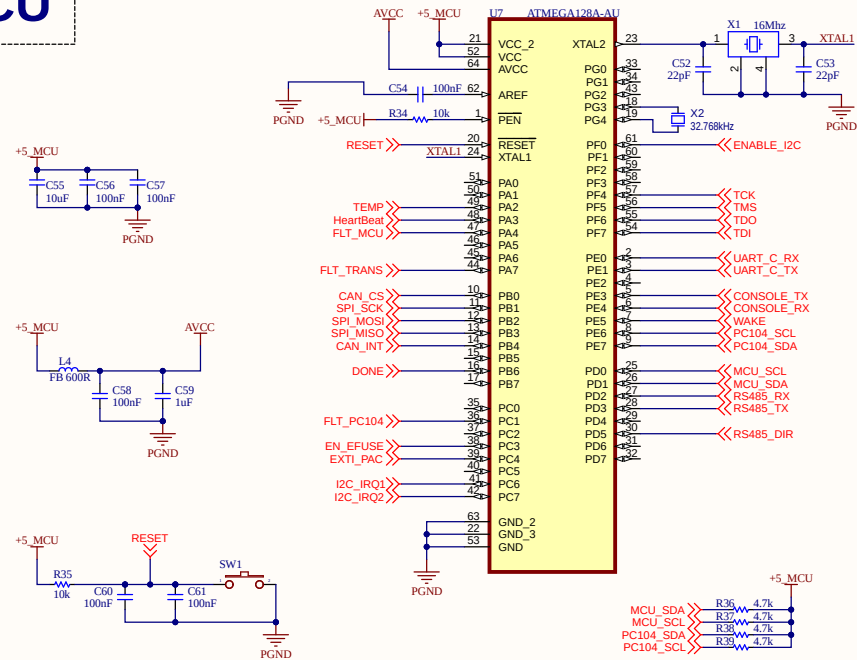
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5-12V BUCK



Project: DAL_PDU	Sub-system: IGRVT-50	
Design by: KIEN HOANG	Sheet title: Buck 5-12V.SchDoc	Sheet 4 of 15
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MCU



Project: DAL_PDU	Sub-system: IGRVT-50	
Design by: KIEN HOANG	Sheet title: MCU.SchDoc	Sheet 5 of 15
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Approved by: VU PHAM	Revision: *	Date: 7/29/2026

CAN

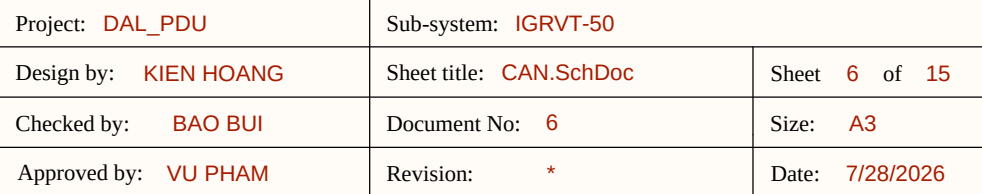
The image displays two circuit diagrams for interfacing a CAN bus with an STM32F407VGT6 microcontroller.

Top Diagram: MCP2515 CAN Transceiver Interfacing

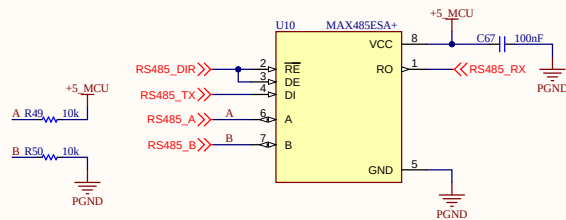
- MCU (U8):** MCP2515-SO.
- Transceiver (U9):** SN65HVD251DR.
- Power Supply:** +5_MCU.
- Decoupling:** C62 (1uF) and C64 (12pF) are connected to PGND.
- Signal Connections:**
 - MCU TXCAN (pin 1) is connected to TXCAN (pin 18) of the transceiver.
 - MCU RXCAN (pin 2) is connected to RXCAN (pin 17) of the transceiver.
 - MCU CLKOUT/SOF (pin 3) is connected to CS (pin 15) of the transceiver.
 - MCU TX0RTS (pin 4) is connected to SO (pin 14) of the transceiver.
 - MCU TX1RTS (pin 5) is connected to SI (pin 13) of the transceiver.
 - MCU TX2RTS (pin 6) is connected to SCK (pin 12) of the transceiver.
 - MCU OSC2 (pin 7) is connected to INT (pin 11) of the transceiver.
 - MCU OSC1 (pin 8) is connected to RX0BF (pin 10) of the transceiver.
 - MCU VSS (pin 9) is connected to VSS (pin 10) of the transceiver.
- Transceiver Pins:**
 - TXCAN (pin 18) is connected to CAN_H (pin 3) of the transceiver.
 - RXCAN (pin 17) is connected to CAN_L (pin 7) of the transceiver.
 - CS (pin 15) is connected to CAN_CS (pin 4) of the transceiver.
 - SO (pin 14) is connected to CAN_INT (pin 5) of the transceiver.
 - SI (pin 13) is connected to CAN_CS (pin 4) of the transceiver.
 - SCK (pin 12) is connected to CAN_CS (pin 4) of the transceiver.
 - INT (pin 11) is connected to CAN_CS (pin 4) of the transceiver.
 - RX0BF (pin 10) is connected to CAN_CS (pin 4) of the transceiver.
 - VSS (pin 10) is connected to CAN_CS (pin 4) of the transceiver.

Bottom Diagram: SN65HVD251DR CAN Transceiver Interfacing

- MCU (U8):** MCP2515-SO.
- Transceiver (U9):** SN65HVD251DR.
- Power Supply:** +5_MCU.
- Decoupling:** C62 (1uF) and C64 (12pF) are connected to PGND.
- Signal Connections:**
 - MCU TXCAN (pin 1) is connected to TXCAN (pin 18) of the transceiver.
 - MCU RXCAN (pin 2) is connected to RXCAN (pin 17) of the transceiver.
 - MCU CLKOUT/SOF (pin 3) is connected to CS (pin 15) of the transceiver.
 - MCU TX0RTS (pin 4) is connected to SO (pin 14) of the transceiver.
 - MCU TX1RTS (pin 5) is connected to SI (pin 13) of the transceiver.
 - MCU TX2RTS (pin 6) is connected to SCK (pin 12) of the transceiver.
 - MCU OSC2 (pin 7) is connected to INT (pin 11) of the transceiver.
 - MCU OSC1 (pin 8) is connected to RX0BF (pin 10) of the transceiver.
 - MCU VSS (pin 9) is connected to VSS (pin 10) of the transceiver.
- Transceiver Pins:**
 - TXCAN (pin 18) is connected to CAN_H (pin 3) of the transceiver.
 - RXCAN (pin 17) is connected to CAN_L (pin 7) of the transceiver.
 - CS (pin 15) is connected to CAN_CS (pin 4) of the transceiver.
 - SO (pin 14) is connected to CAN_INT (pin 5) of the transceiver.
 - SI (pin 13) is connected to CAN_CS (pin 4) of the transceiver.
 - SCK (pin 12) is connected to CAN_CS (pin 4) of the transceiver.
 - INT (pin 11) is connected to CAN_CS (pin 4) of the transceiver.
 - RX0BF (pin 10) is connected to CAN_CS (pin 4) of the transceiver.
 - VSS (pin 10) is connected to CAN_CS (pin 4) of the transceiver.

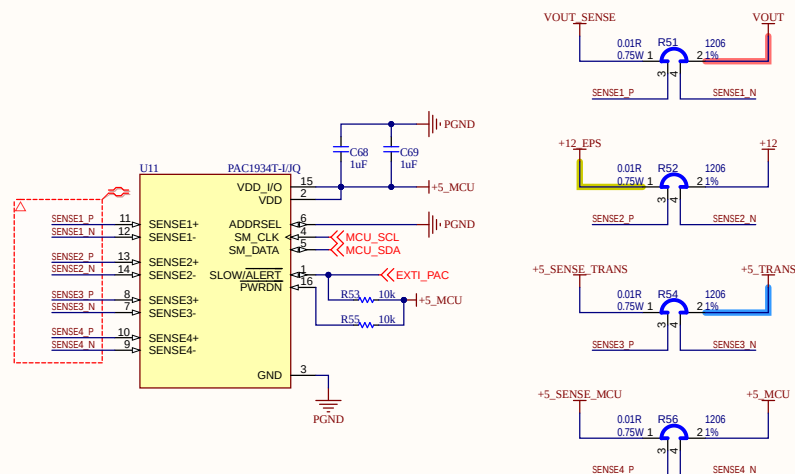


RS485



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Design by: KIEN HOANG	Sheet title: RS485.SchDoc	Sheet 7 of 15
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POWER MONITORING



Project: DAL_PDU

Sub-system: IGRVT-50

Design by: KIEN HOANG

Sheet title: PowerMonitor.SchDoc

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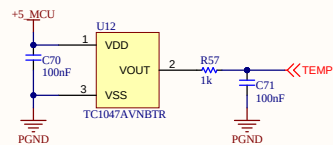
Size: A3

Approved by: VU PHAM

Revision: *

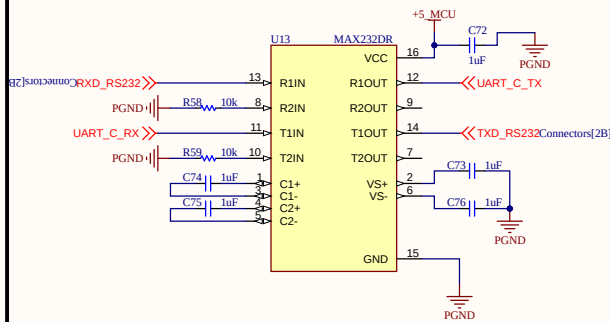
Date: 7/28/2026

TEMPERATURE SENSOR



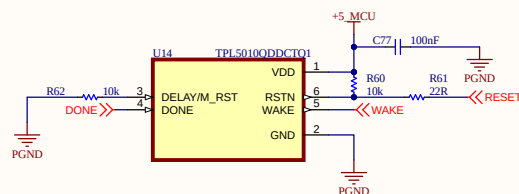
Project: DAL_PDU	Sub-system: IGRVT-50	
Design by: KIEN HOANG	Sheet title: TemperatureSensor.SchDoc	Sheet 9 of 15
Checked by: BAO BUI	Document No: 9	Size: A3
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MAX232



Project: DAL_PDU	Sub-system: IGRVT-50	
Design by: KIEN HOANG	Sheet title: MAX232.SchDoc	Sheet 10 of 15
Checked by: BAO BUI	Document No: 10	Size: A3
Approved by: VU PHAM	Revision: *	Date: 7/28/2026

WATCHDOG



Project: DAL_PDU

Sub-system: IGRVT-50

Design by: KIEN HOANG

Sheet title: WatchDog.SchDoc

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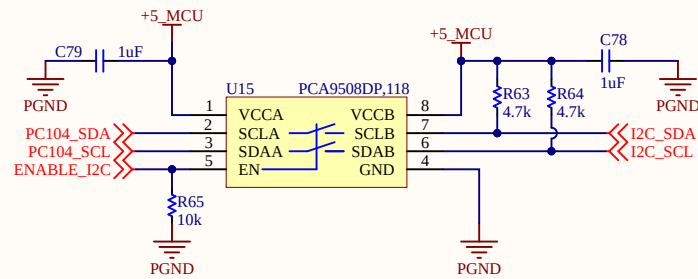
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Approved by: VU PHAM

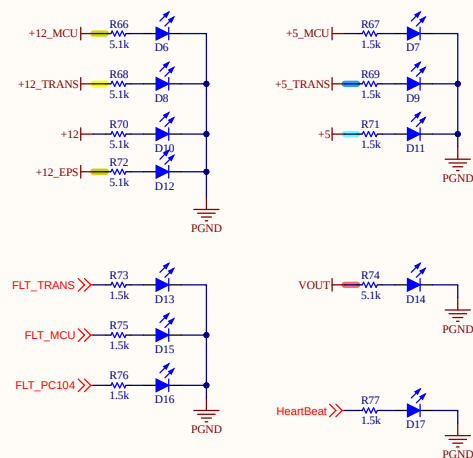
Revision: *

Date: 7/28/2026

WATCHDOG

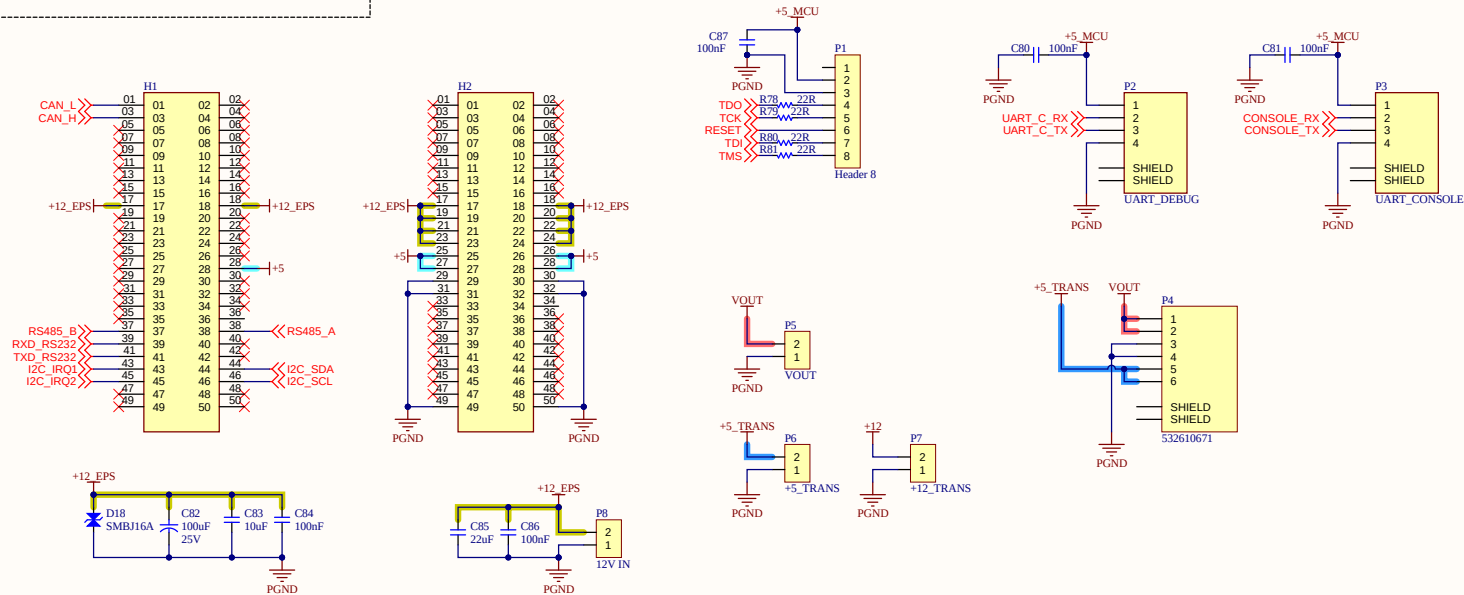


LED



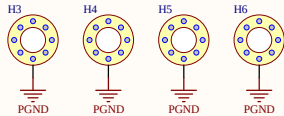
Project: DAL_PDU	Sub-system: IGRVT-50	
Design by: KIEN HOANG	Sheet title: LED.SchDoc	Sheet 13 of 15
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CONNECTOR



Project: DAL_PDU	Sub-system: IGRVT-50	
Design by: KIEN HOANG	Sheet title: Connectors.SchDoc	Sheet 14 of 15
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HOLES



Project: DAL_PDU	Sub-system: IGRVT-50	
Design by: KIEN HOANG	Sheet title: Mounting Holes.SchDoc	Sheet 15 of 15
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Approved by: VU PHAM	Revision: *	Date: 7/28/2026